

THE UNIVERSITY *of* EDINBURGH
School of Engineering

Computational Fluid Dynamics 5

CFD Simulation of a High Speed Shinkansen in the Channel Tunnel

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December 26, 2025

1 Introduction

The purpose of this report is to determine the impact of operating the E5/H5 class Shinkansen in the Channel Tunnel at a speed of 313 km/h, significantly above the current 160 km/h speed limit. Using Star-CCM+, the fluid flow and forces on the train and tunnel walls, including side passages, are determined. These values will allow the client to assess whether the current geometry and material properties of the tunnel are suitable for Shinkansen operation.

2 Train and Tunnel Design

The 3D model provided by the client consists of two carriages, the leading carriage with a standard class passenger carriage behind, equipped with a PS208 pantograph. In Japan, the E5 and H5 Shinkansen trains consist of 10 carriages with two end cars [1]. To accurately simulate the wake but to avoid unnecessary computational cost, two end cars with two intermediate cars are simulated.

Details of the train that cannot be modelled without an extremely fine mesh, and don't have an impact on the overall fluid motion, were removed. Included were the windscreen wipers, vents on the side of the train, and the pantograph, as its aerodynamic influence is localised and does not significantly affect the bulk flow characteristics or quantities such as pressure increases and wall forces. Subsequently, the aero fins are removed, as their purpose is to alter the flow generated by the pantograph. Note that the removal of these features will reduce the total drag force coefficient C_D of the train.

The model provided is a scale model. The width was scaled up 68 times, determined by comparing the model gauge to the standard gauge of the Channel Tunnel and Shinkansen network. The length was scaled up 90 times and height 83 times, based on the Shinkansen carriage dimensions [1].

Literature detailing the Channel Tunnel internal geometry is scarce, but the tunnel diameter is known to be 7.6 m [2]. The client has provided dimensions for the side passages: 4 m bore diameter, 2 m deep, with a 1.5 m wide flat bottom that is 1.5 m from the tunnel floor. As per the running directions, the cross passage is on the right-hand side. The Channel Tunnel's 2 m diameter piston relief ducts are not be modelled as the client needs to understand the unrelieved pressure build-up in the tunnel [2].

To accommodate the Shinkansen, modifications to the tunnel floor are required. The Channel Tunnel uses a UIC GC loading gauge of width 3,150 mm, while the Shinkansen is of width 3,350 mm [1] [3]. Thus, the width of the tunnel floor is 3400 mm, with 100 mm high and 200 mm wide steps before 0.9 m wide walkways on either side, 0.4 m above the tunnel floor. The rail sleepers are modelled as a 2.72 m by 30 mm block along the length of the tunnel, as they will appear from the perspective of the high-speed Shinkansen.

Figure 1 shows the train and tunnel model, with key dimensions. The upstream and downstream tunnel lengths will be determined by the preliminary simulations, but are initially set at 3 pipe diameters (22.8 m) upstream and 10 pipe diameters (76 m) downstream.

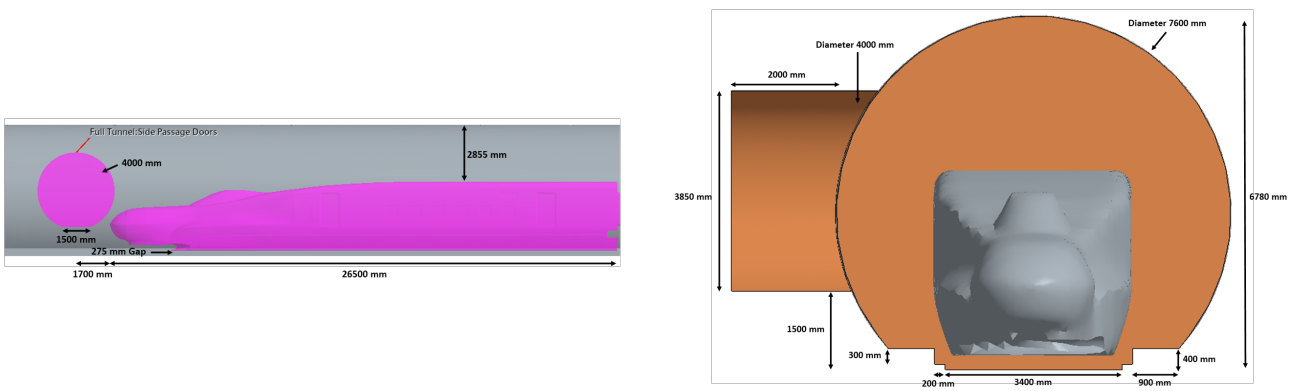


Figure 1: Model of four carriage Shinkansen train in the Channel Tunnel

3 Selection of Physics Models

The simulation solvers require several environmental parameters to solve, which are specific to the Channel Tunnel: pressure and temperature. The tunnel's cooling system maintains a temperature of 25°C, although the permissible operating range is 15-35°C, so 15°C is used as it is denser [1] [4]. The tunnels are maintained at ambient pressure. Although the midpoint of the tunnel is approximately 90 m below sea level, it is not considered hydrostatic due to the piston effect of trains moving through the tunnel and ventilation systems. The hydrostatic pressure addition at the midpoint depth would be 1 kPa, which is negligible.

The Shinkansen will be travelling at a speed of 313 km/h, or 86.94 m/s, a Mach number of 0.26, below the 0.3 threshold to satisfy the incompressibility condition. However, as the train moves through the tunnel, the existing air is severely compressed and accelerated between the train and the tunnel walls. Therefore, a compressible solver is used.

Air is modelled as an ideal gas so that density varies consistently with local pressure (and temperature), using $\rho = p/(RT)$. This is appropriate because tunnel air remains close to atmospheric pressure and ambient temperature, where real-gas effects are negligible. The interactions between the train and tunnel produce pressure waves that are more accurately represented when density is allowed to respond to pressure changes rather than being fixed as a constant. The gradient of the Channel Tunnel ranges from 0.2 to 1.1%, with the most extreme values at either end of the tunnel [5]. Therefore, the tunnel will be assumed to be horizontally level, and gravity is not modelled, as there are no vertical pressure gradients.

The Reynolds number is calculated using the velocity U = 86.94 m/s, tunnel diameter D = 7.6 m and approximate dynamic viscosity 1.56 $\times 10^{-5}$ m^2/s. Re = 4.24 $\times 10^7$, therefore the flow is turbulent. The SST k-ω turbulent model was chosen. The k-ϵ model performs well for external flows and free streams, but is less accurate in regions of strong pressure gradients and flow curvature, which are present in this study. The k-ω model performs better for internal flows (such as tunnels) and separated regions. The SST (Shear Stress Transport) model combines both models.

4 Simulation 1

4.1 Simulation Design and Workflow Strategy

The first simulation is the train passing through the midpoint of the tunnel without a cross-passage. This means that the tunnel cross-section is identical along the length of the tunnel, which can be taken advantage of for this simulation. In real life, the train will be moving at 313 km/h relative to all features in the tunnel. To simulate this in the tunnel, the frame of reference is flipped, such that the train surface can be set to be stationary, while the inlet flow can be set to 313 km/h, as well as the tunnel surfaces. With no cross passage, there is no time-dependent geometric forcing, and the quantities of interest (forcing on walls and train) are controlled by steady mean flow. As the flow is symmetric, a half-tunnel model is used.

A preliminary study was set up on a coarse grid, used to inform the refinement required. On a refined mesh, a grid refinement study will be conducted, and a final simulation with further refinement. The integral quantity selected for monitoring and grid refinement is the drag coefficient of the train, C_D. It is a non-dimensional, globally integrated measure of the train's aerodynamic performance combining pressure effects and viscous forces. Therefore, it is well-suited for assessing grid convergence independence.

4.2 Preliminary Simulation

As both the train and tunnel are symmetric along the length, the cross passage is not included; only half of the domain is required, which will halve computational requirements. With a train length of 103 m, and upstream lengths of 3 (22.8 m) and 10 (76 m) pipe diameters, the tunnel modelled is 201.8 metres long. By conducting a study later with a downstream length of 50 pipe diameters (505.8 m) with section planes used to measure static pressure, it was found that the optimum downstream length is 35 pipe diameters (391.8 m), where the pressure difference between two points is under 1%, so pressure has recovered.

Mesh control values were: base size 1 m, target size 50% of base, minimum size 2% of base, cell growth rate 1.2 (reduced from the default 1.3). The number of prism layers was increased to 15 for better near-wall accuracy, with the total thickness set to 0.1 m. Prism layers were applied to the train and tunnel walls.

The preliminary simulation produced a drag coefficient of 1.07303. With the unsteady solver $C_D = 1.07768$, a 0.4% increase, with high residuals. This indicates any unsteady flow features make a negligible contribution to the drag, and the flow is effectively quasi-steady, so the steady solver is appropriate.

4.3 Mesh Refinement

4.3.1 y^+ Values

From the preliminary simulation, the wall y^+ values were plotted across the length of the train and along the length of the tunnel. Train y^+ values ranged from 3 to 9, and the wall values ranged from 4 to 12. For the k-ω turbulence model, resolution of the viscous sublayer is required, so y^+ values should be below 1.

The y^+ values depend on the number of prism layers and total thickness, which should cover the entire boundary layer thickness, which is dependent on the Reynolds number. The boundary layer along the train's length is calculated as follows, where the density is obtained from the simulation's initial conditions:

$$Re_L = \frac{\rho U L}{\mu} = \frac{1.18 \times 86.94 \times 103}{1.86 \times 10^{-5}} = 569.63 \times 10^6 \quad \Rightarrow \quad \delta = \frac{0.38L}{Re_L^{\frac{1}{5}}} = \frac{0.38 \times 103}{(569.63 \times 10^6)^{\frac{1}{5}}} \approx 0.7 \text{ m} \quad (1)$$

It is recommended that for $y^+ < 1$ the number of prism layers N be no less than 25, therefore $N = 30$ is chosen [6]. Equation 2 is used to calculate the height of the first prism layer, where τ_w = 6.19 Pa, obtained using a surface average report in Star-CCM+. Equation 3 is used to determine the growth rate, G, which is found to be 1.43. Note that the first prism layer height is 6.88 microns, which is very small; therefore, for full near-wall resolution with the k-ω solver, it is computationally expensive.

$$y_H = \frac{y^+ \nu}{u_\tau} = \frac{y^+ \mu}{\sqrt{\tau_w \rho}} = \frac{1 \times 1.86 \times 10^{-5}}{\sqrt{6.19 \times 1.18}} = 6.88 \times 10^{-6} \text{ m} \quad (2) \qquad \delta = y_H \frac{1 - G^N}{1 - G} \Rightarrow G = 1.43 \quad (3)$$

The prism layers on the tunnel walls were less straightforward to calculate, as the velocity of the air relative to the tunnel is zero before the train, before being accelerated as the cross-sectional area available decreased. It was found that a total thickness of 0.6 m was sufficient to cover the boundary layer, with $N = 30$ and $G = 1.4$ to keep $y^+ < 1$ along the length of the tunnel. The tunnel floor is not of interest, as due to the lack of train wheels and high level of refinement, the solution will not be accurate there.

4.3.2 Areas of Refinement

From the preliminary simulation, areas of high pressure and velocity gradients were identified at the front and the end of the train, as well as the spacing between the carriages. Refinement was done using volumetric controls with cylinder parts along the tunnel. The front and end refinement cylinders were the same radius as the tunnel to capture the pressure and velocity gradients through to the tunnel walls. The nose refinement was 16 m long, and the end refinement was 30 m long to capture the wake. The carriage spacing refinement blocks, 0.75 metres long, were added. Cell size was set to 10% of the base.

4.3.3 Grid Refinement Study

To determine if the solution is mesh independent, a grid convergence study was conducted. To do this, the drag coefficient was recorded on four different mesh sizes with the most recent refinement. As the preliminary simulation took more computational time to solve than predicted, coarser grids were tested rather than finer. The base was doubled for each grid. The average cell spacing Δx was derived by taking the cube root of the mesh volume divided by the number of cells. The results are shown in Tables 1 and 2 and Figure 2, produced using a Python script [7]. The size of the prism layers was kept absolute.

Table 1: Grid parameters and drag coefficient

Grid	Base size	Cells	Δx	**Ratio**	C_D
1	1 m	1,004,263	0.1416	1.4	0.7301
2	2 m	367,904	0.1980	1.3	0.8772
3	4 m	168,026	0.2571	1.4	1.2147
4	8 m	62,786	0.3569	–	1.7028

Table 2: Grid Convergence Index (GCI)

Grid Pair	GCI	GCI Ratio	Converged
1 2	0.1946	1.201	No
2 3	0.3716		
2 3	0.3716	2.197	No
3 4	0.3880		

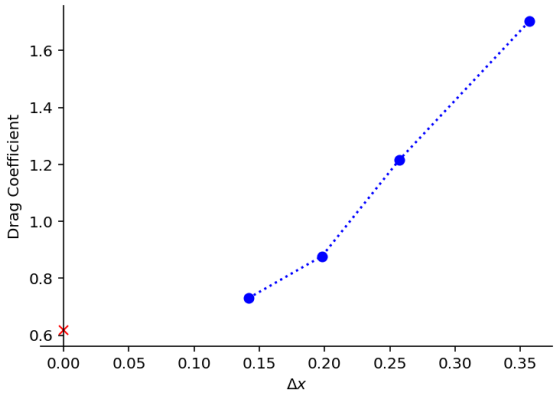


Figure 2: Drag coefficient grid convergence

The drag coefficient decreases consistently with decreasing grid spacing, indicating consistent convergence behaviour. The final order of convergence was $p = 2.48$, consistent with near second-order spatial accuracy. The GCI of the two finest grids was 19.46%, showing that while there are convergence trends, numerical uncertainty remains and further refinement is required to achieve grid independence. This is not possible with the hardware provided by the client.

4.4 Production Simulation

As a base size of 1 m was found to be the limit that the hardware can solve within a reasonable time frame, the 1 m base size was used for the final production run. An additional surface control was applied to the train surface, setting target and minimum cell size to 10% of the base size, matching the volume refinement.

Figures 3, 4, and 5 show contour plots of the Mach number, pressure, and density distributions using plane sections along the midplane of the train and a horizontal plane at the height of the train's nose. Figure 3 shows that there is a large area of wake created behind the train at either side, while there is virtually no wake along the plane of symmetry, with only recirculation underneath the rear nose, as is seen in literature [8] [9]. Figures 6 and 7 shows recirculation within the wake. Flow is accelerated between the train and the wall. A very thin boundary layer is created on the wall as the velocity differential is small, and a much larger one is produced by the train. The flow in this boundary layer is not moved by the train. It can also be seen that there is a stagnation point at the nose of the train, where the highest pressure (8.53 kPa above gauge) and density (1.27 kg/m^3) occur. Figure 4 shows that the minimum pressure, -7 kPa, occurs where the maximum flow velocity occurs near the rear of the train. The flow speed here is 0.31 Mach. Behind the train, the pressure and density recover.

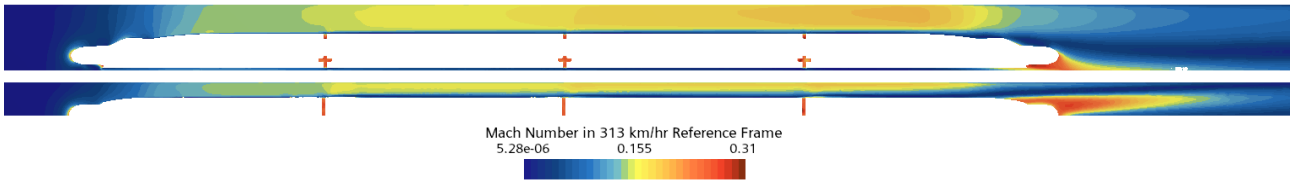


Figure 3: Mach number contour plots in vertical (top) and horizontal (bottom) cross-sections

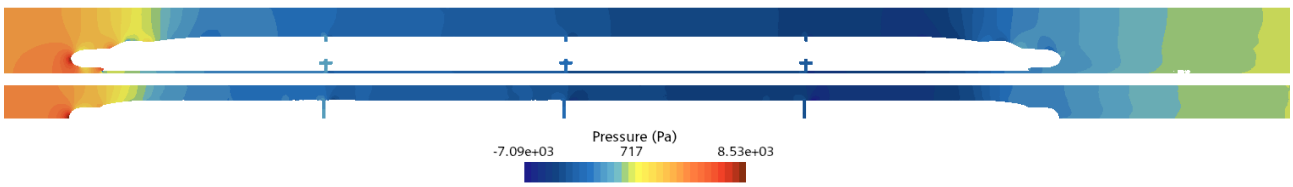


Figure 4: Gauge pressure contour plots in vertical (top) and horizontal (bottom) cross-sections

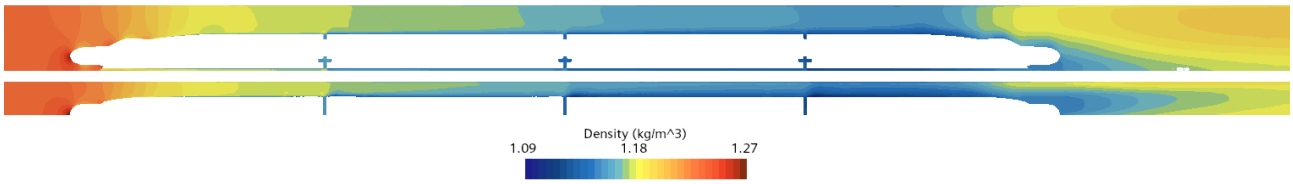


Figure 5: Density contour plots in vertical (top) and horizontal (bottom) cross-sections

Figure 8 shows the pressure along the train and tunnel, as the train pushes the air along. By the method of force and surface integral monitor plots, the drag and lift forces on the 4-carriage train were 33.56 kN and 10.86 kN, while the side force was 1,489 kN. The load on the tunnel walls was -3.972 MN, meaning that the net force was pulling inwards due to the effect of low pressure. The maximum pressure was 1.834 kPa, and the minimum was -5.793 kPa.

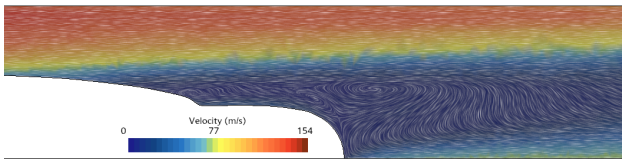


Figure 6: Vector scene showing wake in a horizontal cross-section, velocity from train's frame of reference

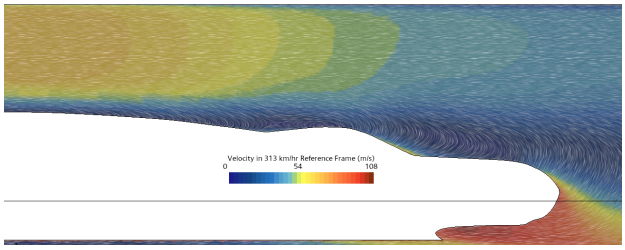


Figure 7: Vector scene showing separation in a vertical cross-section

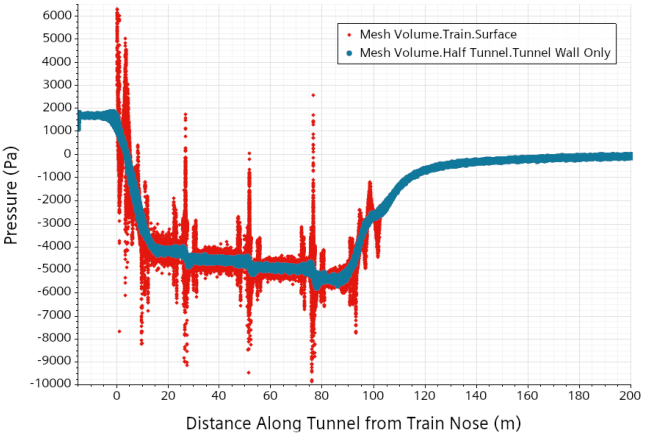


Figure 8: Pressure distribution on tunnel (blue) and train (red), highest at the train nose before decreasing along the length of the train before recovery beyond the wake region

5 Simulation 2

The objective of the second simulation is to analyse the forces that act on the side tunnel passage walls and, importantly, the back face, which will be a door. To simulate this behaviour without creating a moving mesh, a similar approach will be used.

The method of simulation begins with a convergent steady-state solution of the tunnel, identical to Simulation 1, but with a full tunnel modelled, as the simulation is unsymmetrical with a side passage on the right-hand side in the running direction. Once converged, the geometry of the side tunnel is added. From Figure 8, the maximum pressure occurs 1.74 m in front of the train, so the centroid of the side passage is placed here. A mesh refinement cylinder is placed with a size 5% of the base, and with prism layers removed due to the recirculation expected. All other mesh and geometry values are carried over from Simulation 1. The solver is then changed to implicit unsteady, with monitor plots set up to measure the shear and compressive forces on all side passages surfaces over time. The flow within the side passage will initially be stationary relative to the wall, before being hit by the pressure wave at the front of the train.

The time step is selected to ensure the CFL number is less than 1, thereby ensuring stability in both space and time. The flow speed used is the maximum observed in simulation 1: 160 m/s. As the simulation deals with compressible flows, the speed of sound, a, is accounted for. The time step chosen is 0.0002 seconds.

$$\Delta t \leq \frac{\text{CFL} \times \Delta x}{u_{max} + a} = \frac{1 \times 0.104}{160 + 340} = 0.000208, \quad t = 0.0002 : \quad \text{CFL} = \frac{(160 + 340) \times 0.0002}{0.104} = 0.96 \quad (4)$$

It was found that this setup created problems as the air within the side passage becoming stagnant and creating recirculation like flow over a cavity, as if there was actually moving flow within the tunnel. To fix this, a velocity inlet and pressure outlet boundary condition were applied to the upstream and downstream faces. Figure 9 shows the shear force on the side walls and the compressive force on passage doors. The maximum compressive force on the doors is 114.3 kN, and occurs at 0.0334 s. Figure 10 shows the pressure variation just after the peak of 30 kPa. Peak shear force on the side passage walls is 198.1 kN. It takes 0.046 seconds for the nose of the train (and peak pressure) to pass the side passage. Peak compressive force on the side passage walls is 339.3 kN, before the boundary conditions were altered. When comparing to material properties, a safety factor should be applied to account for simulation errors.

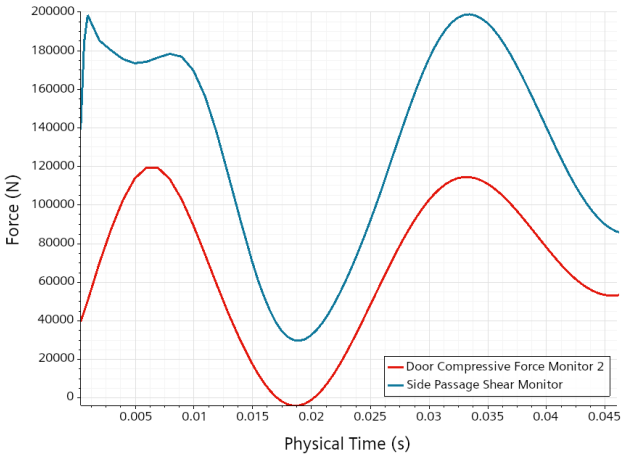


Figure 9: Door compressive force and side passage shear force over time taken for pressure to pass

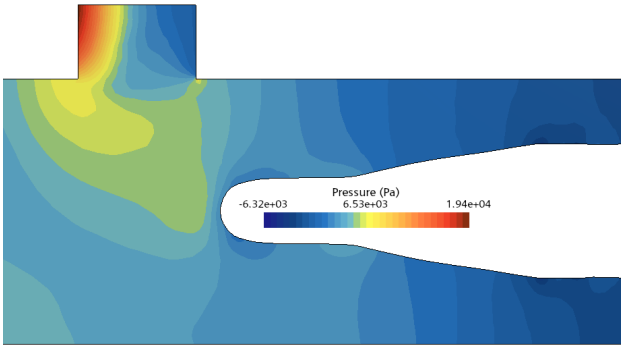


Figure 10: Pressure distribution in side passage at 0.04 seconds, post pressure peak. Peak pressure occurs between the passage doors and side facing the train

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